ELEG 3124

Assignments for Lab 0 Matlab Basics

Notes:

1. Most of the lab computers have Matlab installed already.
2. If your computer does not have Matlab, you can access Matlab by using http://vlab.uark.edu
3. Before working on the homework, please watch the 5 videos posted under Lab 0 on the course website: http://comp.uark.edu/~wuj/teaching/eleg3124/eleg3124.html
4. The "help" command is your best friend. Whenever you are not clear about the function of a Matlab command, use "help". For example, the following command

   ```
   >>help linspace
   ```

   Will display the detailed information of the Matlab function linspace.
5. Please save everything in a script file, and submit your ".m" script file. The TA will directly execute your .m file in Matlab to check for results.

Assignments

1. Following the example, write the set of Matlab commands required to perform the following operations in Matlab and give the final answer:

   Example: write the Matlab commands to perform the following operations

   $x = 25, y = 2, z = \sqrt{x} + 3y^2$

   Answer:

   ```
   x = 25
   % if a line is ended with ";", then the result will not be
   displayed
   y = 2;
   % please note the multiplication operator * cannot be omitted
   % if you type sqrt(x)+3y^2, matlab will give you an error
   z = sqrt(x)+3*y^2
   ```

   Questions:

   a. $x = 9, y = 4, z = 2x^2 - \sqrt{y}$
   b. $x = 6.2832, y = \pi, z = \sin(x) + \cos(y)$
   c. $x = [1, 2.4, 6.8, 8, 10], y = [0.7, 4, 2, 5.4, 9], z = \log(x) + \log(2y)$
   d. $x = [4, 6, 9,12], y = [1, 2,1,2], z = x \times y$

Hint: please use the element wise multiplication command ".*" (a dot followed by a star). Use "help .*" to get more information.

e. $x = [4, 6, 9, 12], y = [1, 2, 1, 2], z = x/y$
Hint: please use the element wise division command "./" (a dot followed by a slash). Use "help ./" to get more information.

f. $x = [1, 2, 3, 4, 5], z = x^2$
Hint: please use the element wise power command ".^" (a dot followed by a hat). Use "help .^" to get more information.

2. Following the example, write the set of Matlab commands to plot y as a function of x

Example: write the Matlab commands to plot x as a function of y

$x = [0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4], y = x^2$

Answer:

```
% generate a vector x, starting at 0 and ending at 4. The step
between two
% adjacent eleemnts is 0.5
% that is: x = [0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4]
x = [0:0.5:4];
y = x.^2;
plot(x, y);
```

Questions:

a. $x = [1,2,3,4], y = x^3$
b. $x = [0: 0.1\pi: 2 * \pi], y = \cos(x)$
c. $x = [0: 0.1: 2 * \pi], y = \cos(x)$
d. $x = [0: 0.1: 2 * \pi], y = e^{2x} + \sin(x^2) + \tan(\sqrt{x})$
e. $x = 0.1: 0.1: 10, y = \cos(x) / \log(x)$

3. Matlab operators. Execute the following commands in Matlab, and understand the results. Use help to get the description of the operators used in the examples.

a.

```
x = [1 5 2 8 9 4 1]
y = [5 2 2 6 0 0 2]
x > y
```

b.

```
x = [1 5 2 8 9 4 1]
y = [5 2 2 6 0 0 2]
x > y
```

c.

```
x = [1 5 2 8 9 4 1]
y = [5 2 2 6 0 0 2]
x == y
```

d.

```
x = [1 5 2 8 9 4 1]
y = [5 2 2 6 0 0 2]
x <= y
```

e.

```
x = [1 5 2 8 9 4 1]
y = [5 2 2 6 0 0 2]
x ~= y
```

f.

```
x = [1 5 2 8 9 4 1]
y = [5 2 2 6 0 0 2]
x ~= y
```

g.

```
x = [2:2:8]
y = [1:4]
x.*y
```

h.

```
x = [2:2:8]
y = [1:4]
x./y
```